



charging operations with real-time convenience and reliability.

With a fully operational prototype, sparXcharge has achieved certification and safety validation. The prototype is designed with high-capacity mobile chargers capable of handling various power needs, ranging from 30kW to 120kW, allowing each van to charge multiple EVs simultaneously. The prototype has been deployed in Trivandrum, demonstrating its potential in real-world scenarios. A full-scale soft launch with ten mobile charging stations is scheduled for Q2 2025, shares Imthias Mohamed.

sparXcharge employs a three-tier strategy to redefine EV charging:

Mobile EV charging. Battery-powered vans equipped with high-capacity chargers are dispatched to customers' locations, eliminating the need for grid dependency.

Fast-charging stations. sparXcharge is establishing its fast-charging stations in urban areas to recharge mobile units and serve the public, creating an additional revenue stream.

IoT-enabled software platform. A comprehensive software platform with IoT integration enables real-time monitoring of charging status, manages logistics, and streamlines customer interactions. This digital platform combines charge management, tracking, and billing, ensuring seamless and efficient service delivery.

The startup serves both B2B and B2C markets with customised EV charging solutions.

"For our B2B clients, we partner with businesses like malls, workplaces, and large organisations to provide on-site charging, ideal for EV fleet operators who benefit from on-demand or scheduled charging, significantly minimising downtime," shares Imthias Mohamed.

"For individual consumers, we offer a subscription model



Imthias Mohamed, Founder and CEO, sparXcharge

with flexible monthly and annual options, allowing convenient, station-free vehicle charging at a predictable cost. Our emergency 'rescue charging' service also supports customers with vehicles stranded due to low battery levels," adds Imthias Mohamed.

Each service operates within a three-kilometre radius, allowing sparXcharge to respond to any request within ten minutes. Currently, Trivandrum is divided into ten such zones, with plans to expand this model to other Indian cities and major highways.

This startup is not engaged in manufacturing. Instead, the company collaborates with OEM partners who handle production based on sparXcharge's customised designs and specifications. This partnership model allows sparXcharge to focus on service innovation and quality control without the logistical complexities of manufacturing. The core of its technology involves a combination of battery management systems and the Internet of Things for real-time monitoring and control.

"We offer subscription packages tailored to different usage frequencies and distances, making them attractive to cost-sensitive customers

who might otherwise prefer home charging. In cities like Trivandrum, we currently charge around 12 rupees per kilowatt-hour and have seen steady daily demand for our services," explains Mohamed.

sparXcharge also offers on-demand top-up charging for occasional users, allowing them to pay per session as needed. Its rescue charging service immediately assists customers with low EV battery levels.

Imthias highlights key challenges in India's EV charging sector, such as limited EV range, a charging infrastructure shortage, and difficulties securing land for charging stations. sparXcharge's mobile model addresses these issues, offering flexible services that bypass the need for permanent infrastructure. The company stays attuned to technological trends, preparing for advancements like high-capacity EV batteries and adapting its services to meet future demands.

The startup collaborates with real estate companies and equipment manufacturers for infrastructure support. Through these partnerships, it gains access to strategic locations for mobile and stationary charging units without incurring high upfront costs. These partnerships also involve a revenue-sharing model, where sparXcharge oversees customer interaction and billing while paying a monthly rental fee for equipment and space usage.

sparXcharge aims to expand to 25 cities by 2030, with 250 mobile charging stations expected to be deployed across India. Initial expansion will focus on southern cities like Bengaluru and Coimbatore, followed by larger metro areas. It also plans to introduce high-capacity chargers for highway applications, accommodating larger EVs and commercial vehicles, such as trucks and buses, which require higher power capacities. **EFY**

—Nitisha Dubey